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TOLERANCE VALUES OF STREAM CADDISFLIES (TRICHOPTERA) IN THE LOWER COLORADO RIVER BASIN, USA

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ABSTRACT—One hundred and four caddisfly species within 42 genera and 17 families were collected from 93 stream sites in the xeric landscape of the lower Colorado River Basin, USA. Species richness showed a significant negative correlation with channel embeddedness. Forestland had higher species richness than grassland, desert, or urban caddisfly assemblages, and fewer caddisfly species occurred in the salt-cedar (*Tamarix*) than in willow-alder (*Salix-Alnus*) and cottonwood-sycamore (*Populus-Plantanus*) riparian communities. Hydroptilidae comprised nearly 35% of the average relative abundance of caddisflies and were generally tolerant of impaired stream environments. *Hydroptila arctica* composed the greatest relative proportion of species. Adjusted specific conductance and channel embeddedness metrics showed close agreement with published tolerance values for caddisfly species listed by the U.S. Environmental Protection Agency; genera were in less agreement. *Amphicosmoecus canax*, *Brachycentrus americanus*, *Culoptila moselyi*, *Glossosoma ventrale*, *Ochrotrichia ildria*, *Psychoglypha schuhi*, and *Ylodes reuteri* were most commonly associated with streams with low salinities and low embeddedness, whereas *Cheumatopsyche enonis*, *Hydropsyche auricolor*, *Hydroptila ajax*, *Hydroptila arctica*, *Neotrichia okopa*, *Smicridea signata*, and *Smicridea fasciatella* reached highest numeric importance in streams with high salinities and embeddedness.

RESUMEN—Ciento cuatro especies de Tricóptera de 42 géneros y 17 familias fueron colectadas de 93 arroyos en ambientes áridos de la cuenca baja del río Colorado, USA. La riqueza de especies mostró una correlación negativa significativa con la sedimentación del lecho. La tierra forestal tuvo más riqueza de especies de Tricóptera que los pastizales, el desierto o centros urbanos, y menos especies de Tricóptera ocurrieron en cedro de la sal (*Tamarix*) que en las comunidades ribereñas de sauce-aliso (*Salix-Alnus*) y álamo-sicamor (*Populus-Plantanus*). Hydroptilidae comprendieron casi 35% de la abundancia relativa promedio de Tricóptera y fueron generalmente tolerantes de ambientes de arroyos dañados. *Hydroptila arctica* tuvo la más alta proporción relativa de las especies. Las medidas para conductividad específica ajustada y para la sedimentación del lecho fueron parecidas a valores de tolerancia para Tricóptera publicados por la U.S. Environmental Protection Agency; las de los géneros estuvieron menos de acuerdo. *Amphicosmoecus canax*, *Brachycentrus americanus*, *Culoptila moselyi*, *Glossosoma ventrale*, *Ochrotrichia ildria*, *Psychoglypha schuhi*, y *Ylodes reuteri* fueron comúnmente asociadas con arroyos con salinidades bajas y sedimentación baja, mientras que *Cheumatopsyche enonis*, *Hydropsyche auricolor*, *Hydroptila ajax*, *Hydroptila arctica*, *Neotrichia okopa*, *Smicridea signata*, y *Smicridea fasciatella* alcanzaron importancia numérica más alta en arroyos con altas salinidades y sedimentación.

Although tolerance values are available for macroinvertebrates for nearly all regions in the USA (Barbour et al., 1999), the lack of such values for the American Southwest demonstrates the scarcity of knowledge on the ecology of macroinvertebrates for the region. The Southwest is a topographically varied, mostly semiarid region, with isolated streams separated by large expanses of xeric landscape.

The lower Colorado River basin (hereafter referred to as the lower basin) is large (about

417,300 km²), especially diverse, and includes 4 major river drainages (Colorado River, Little Colorado River, Salt River, and Gila River), with many of their channels traversing semi-arid landscapes. The region encompasses 2 physiographic provinces (Basin and Range and Colorado Plateaus), 3 freshwater ecoregions (Vegas-Virgin, Colorado, and Gila rivers), and 6 terrestrial ecoregions (Arizona Mountains Forest, Madrean Sky Islands Montane Forest, Colorado Plateau Shrublands, and the Chihu-

ahuan, Mojave, and Sonoran deserts) (Ricketts et al., 1999; Abell et al., 2000). Much of the landscape throughout the lower basin receives heavy use from mining, grazing, and agriculture, principally cotton, all of which contribute to increased channel embeddedness and secondary salinization (Blinn and Poff, 2005).

While the lower basin includes a variety of aquatic environments, only a few papers have dealt with the distribution and ecology of caddisflies in the region. Moulton et al. (1994) published on the caddisflies in north-central Arizona, and Houghton (2001) published on the caddisflies of the Apache National Forest in eastern Arizona. Both studies reported new distributional records for the region and several undescribed species. Blinn and Ruiter (2005) conducted a preliminary study on caddisfly distribution and water quality conditions in Arizona, and Spindler (1996) correlated caddisfly genera with ecoregions at about 100 reference sites throughout the state. Less extensive studies have been conducted on caddisfly genera in Oak Creek (Scott, 1982; Dehoney and Gaud, 1983), Beaver Creek (Oberlin, 1995), and the confluences of tributaries into the Colorado River in Grand Canyon National Park (Hofnecht, 1981; Oberlin et al., 1999).

Several workers have addressed the value of stream caddisflies in other regions in North America as a measure of the biological integrity of streams (Johnson et al., 1993; Resh et al., 1996; Hawkins and Vinson, 2000; Black et al., 2004). Streams in the lower Colorado River basin have great potential for testing the efficacy of caddisflies as indicators of water quality because of the variety of ecological conditions that exist. Although tolerance ranges for caddisflies are available for other regions in the USA, most of these species do not occur in the lower basin.

Collections were made at 90 stream sites throughout Arizona and 3 adjacent sites in Utah to determine the distribution and com-

position of caddisfly communities in the lower basin (Fig. 1). Our study provides an updated list of caddisfly species for the region and the role of specific conductance (salinity) and channel embeddedness on the distribution of these caddisfly species. Tolerance values, based on channel embeddedness and specific conductance for caddisfly species and genera, were adjusted for comparison with published tolerance values listed by the U.S. Environmental Protection Agency (USEPA).

METHODS—Field Collections—We collected physico-chemical information and caddisflies from 93 stream sites throughout the lower Colorado River basin during the summers (June through September) of 2002 and 2003 (Fig. 1). Specific conductance (at 25°C) of water was determined with a conductance meter (Radiometer, Copenhagen, Denmark; Model CDM2e) at each collection site. Channel embeddedness was determined at each site following Platts et al. (1983) and MacDonald et al. (1991).

Riparian vegetation was grouped into the following 3 numerical categories: willow-alder (*Salix-Alnus*) = 1, cottonwood-sycamore (*Populus-Plantanus*) = 2, and the invasive salt-cedar (*Tamarix*) = 3. Land use within catchments was grouped into the following 4 numerical categories: forest = 1, grassland = 2, desert = 3, and urban = 4 at each site, following Ricketts et al. (1999).

Caddisfly larvae and pupae were handpicked from stream vegetation, rocks, and woody debris for approximately 30 min at each site and preserved in 70% ethanol. Caddisfly adults were collected with a vertical, 30-cm, 8-watt, portable, ultraviolet light over a 19-L, white porcelain bucket. Light traps were placed near the stream 1 to 2 h after sunset. Collections were placed in 70% ethanol. The number of adults of each caddisfly species was recorded from each site, and the relative proportion of each species was determined for each site. Functional groups for caddisfly species were obtained from Merritt and Cummins (1996) and Wiggins (1998).

A metric index for specific conductance (SCI) was used to examine the distribution of caddisfly taxa along a specific conductance (salinity) gradient from the equation previously used by Blinn (1993) and Blinn and Bailey (2001) for diatoms:

$$SCI_x = \frac{\sum_{i=1}^{N_x} [\log_{10}(RA_i \times 100)] (\text{specific conductance, } \mu S/cm)}{N_x}$$

where RA = relative abundance (percent) of species "x" at a given stream site, and N = number of

streams in which species "x" occurred. The SCI provided a relative numerical value for each caddisfly

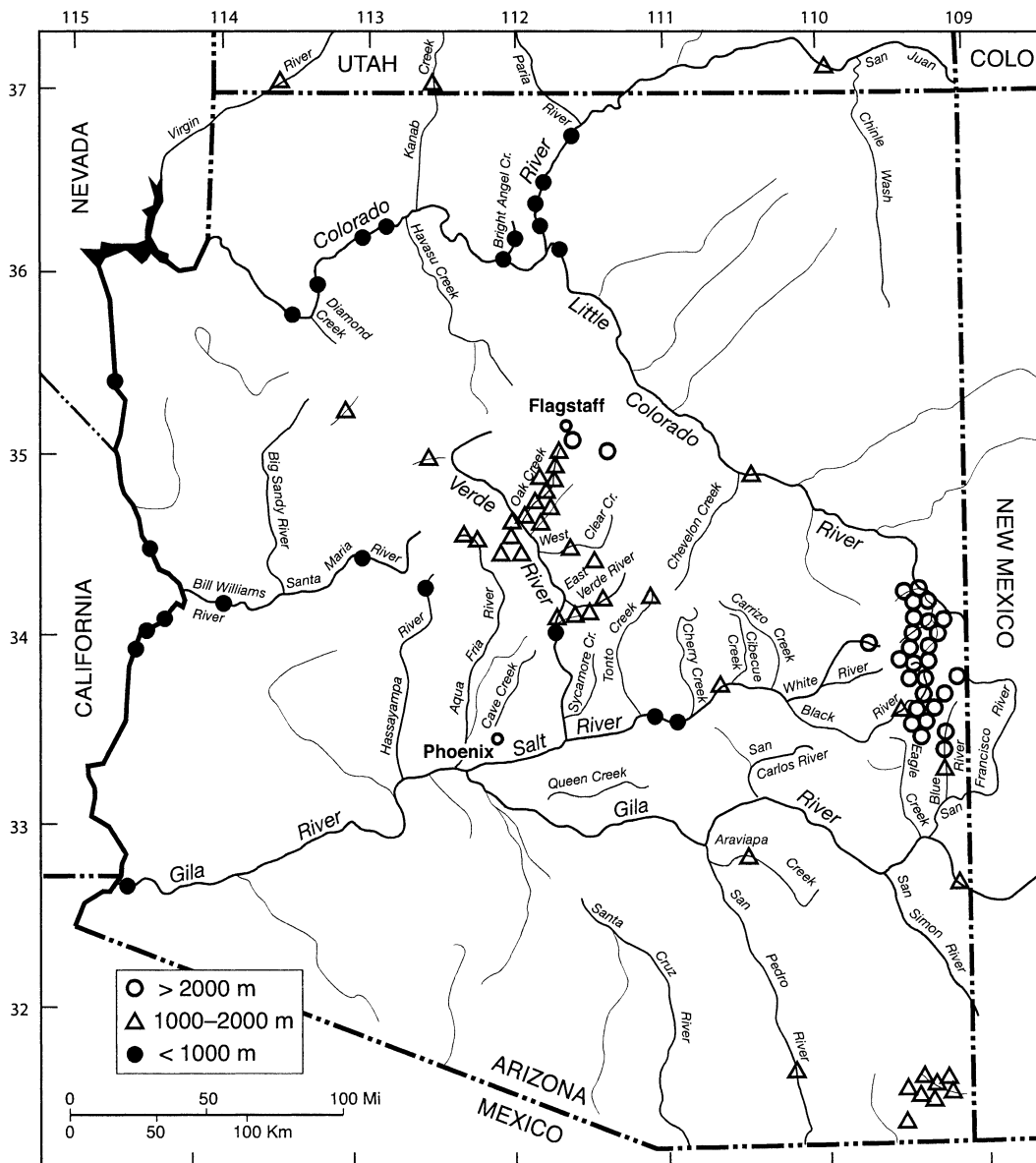


FIG. 1—Locations of 93 stream sites sampled for caddisflies (Trichoptera) in the lower Colorado River Basin, USA, during 2002 and 2003.

species and evaluated the response of various caddisfly taxa to specific conductance. A similar index was calculated for channel embeddedness (EMBI). Siltation of cobbles and salinization of water are 2 of the most influential stressors for stream biota in semiarid and arid regions (Williams, 1987; Laws, 2000). We adjusted the SCI and EMBI relative numerical scores to a scale of 0 to 10 to compare with values from the USEPA reported from other regions in the USA (Barbour et al., 1999).

Correlations between various paired combinations of the relative frequency of caddisfly taxa and environmental determinants were calculated on unnormalized data by using Pearson's product-moment correlation coefficient. Significant differences between paired measurements were obtained from Bonferroni-adjusted alpha values of 0.05. Multivariate analyses of variance (MANOVA) also were used to determine the relationships among abiotic variables (elevation, specific conductance, and embed-

dedness), species richness, and grouped land use and riparian vegetation with Tukey HSD multiple comparisons. All analyses were conducted with SYSTAT software (version 5.1; Wilkinson, 1989).

RESULTS—The 93 stream sites ranged in elevation from 64 to 2,797 m (Table 1). About 40% of the stream sites had riparian communities dominated by willow-alder, about 25% by cottonwood-sycamore, and 35% by salt-cedar. Elevation showed a significant ($P < 0.01$) negative correlation with specific conductance ($r = -0.54$) and channel embeddedness ($r = -0.63$). Specific conductance ranged from 0.01 to 5.23 mS/cm, while channel embeddedness ranged from 5 to 100% (Table 1). Specific conductance had a significant positive correlation ($r = 0.55$, $P < 0.001$) with channel embeddedness.

With respect to land use, 43% of the drainages were in forests, 24% in grasslands, 27% in semideserts, and 6% were in urban areas. Land-use categories were significantly correlated with specific conductance ($F = 15.4$, $P < 0.001$) and channel embeddedness ($F = 5.4$, $P < 0.001$). Tukey multiple comparisons for land use showed significantly ($P < 0.05$) lower specific conductance values for forest compared to other land use categories. Multiple comparisons showed that urban catchments had significantly ($P < 0.05$) higher embeddedness than other land-use categories. There was an increase in specific conductance and channel embeddedness from forest to desert categories. Riparian vegetation was overall significantly ($F = 27.9$, $P < 0.001$) different between land-use categories, with willow-alder communities most common in forests. All multiple comparisons were significantly ($P < 0.05$) different except those between urban:grassland and urban:desert land-use pairs.

Caddisfly Ecology—We identified 104 caddisfly species within 42 genera and 17 families from the 93 stream sites. Species richness showed a significant negative relationship to channel embeddedness ($F = 10.5$, $P = 0.004$; $r = -0.61$). A significant interaction ($F = 4.3$; $P < 0.0001$) occurred between specific conductance and channel embeddedness.

Elevation was an important determinant of caddisfly community composition and species richness. Low-desert streams (<1,500 m) had higher channel embeddedness and salinity and

fewer caddisfly species compared to high-elevation sites. Stream sites above 2,000 m averaged over 3-fold more species than sites below 1,000 m.

Land-use categories showed a significant relationship to species richness ($F = 20.9$, $P < 0.001$). Tukey multiple comparisons showed significant ($P < 0.05$) differences between forest:desert and grassland:desert land-use pairs. Caddisfly species richness averaged 6.6 ($SE \pm 0.9$) taxa in sites with the invasive salt-cedar, compared to 15.6 ($SE \pm 1.1$) in sites with native alder-willow and cottonwood-sycamore riparian vegetation. Also, the LCR Eagar and LCR Springerville sites had less than 12 species, compared to over 20 species at undisturbed sites at similar elevations (Table 1). Urban streams showed a similar reduction in caddisfly species richness with 3-fold to 5-fold more taxa at nonurban sites at similar elevations.

The family Hydroptilidae had the most species with 26, and *Hydroptila arctia* had the highest average relative abundance at 14% ($SE \pm 2\%$) throughout the 93 sites (Table 2). Other numerically important hydroptilids included *Hydroptila ajax*, *H. icona*, *Ithytrichia mexicana*, *Ochrotrichia dactylophora*, *O. logana*, *O. lomenta*, and *O. stylata*. Additional taxa common throughout the lower basin included members of the Hydropsychidae (*Ceratopsyche osleri*, *Cheumatopsyche arizonensis*, *C. enonis*, *C. pinula*, *Hydropsyche auricolor*, and *H. occidentalis*), and *Atopsyche sperryi*, *Chimarra utahensis*, *Gumaga griseola*, *Helicopsyche borealis*, and *Hesperophylax occidentalis* (Table 2). These 19 taxa comprised nearly 60% of the average relative abundance of caddisfly species collected across the lower basin (Table 2). *Ceratopsyche osleri*, *C. arizonensis*, *H. borealis*, *H. occidentalis*, and *O. dactylophora* occurred in at least 30% of the 93 stream sites, and *H. arctia* occurred in 59% of the stream sites (Table 2).

Some caddisfly species, such as *Culoptila cantha*, *Hydroptila rono*, *I. mexicana*, *Lepidostoma knulli*, *L. unicolor*, and *Oecetis disjuncta*, showed high sensitivity to channel embeddedness, whereas others (*C. enonis*, *Smicridea fasciatella*, and *Zumatrichia notosa*) occurred in streams with relatively high channel embeddedness (Table 2). *Hydroptila icona* showed a positive relationship to specific conductance ($r = 0.51$, $P = 0.03$).

The specific conductance (SCI) and embed-

TABLE 1—Location, elevation, specific conductance, channel embeddedness, and caddisfly species richness at stream sites in the Lower Colorado River Basin, USA. Sites are listed in descending order of elevation. Kanab Creek, San Jan, and Virgin river sites are in Utah near the Arizona border. RK = river kilometers.

Stream site	Location	Elevation (m)	Specific conductance (mS/ cm)	Em- bed- ness (%)	Spe- cies rich- ness
Little Colorado River, East Fork	33°55.9'N, 109°29.2'W	2,797	0.054	10	11
Little Colorado River, Sheeps Crossing	33°57.6'N, 109°30.5'W	2,757	0.021	5	21
Black River, Upper West Fork	33°53.4'N, 109°27.6'W	2,641	0.052	5	16
Ackre Lake tributary	33°36.9'N, 109°20.6'W	2,601	0.056	5	8
Little Colorado River, East Fork, Greer	33°59.5'N, 109°27.3'W	2,600	0.112	5	21
K.P. Cienega Creek	33°34.5'N, 109°21.3'W	2,600	0.065	5	19
Fish Creek, FSR24	33°38.2'N, 109°20.5'W	2,556	0.092	5	15
Government Spring	33°59.6'N, 109°27.9'W	2,541	0.116	5	9
Little Colorado River, West Fork, Greer	33°59.5'N, 109°27.9'W	2,541	0.068	5	16
Little Colorado River, Greer	34°00'N, 109°27.4'W	2,539	1.101	5	20
Boneyard Creek	33°50'N, 109°18'W	2,526	0.161	5	21
Black River, 3-forks	33°51.3'N, 109°18.9'W	2,525	0.015	15	37
Halls Creek, Greer	34°03.7'N, 109°27.8'W	2,500	0.019	5	13
Rosey Creek, Greer	34°03.0'N, 109°27.5'W	2,495	1.015	5	26
Benny Creek, Greer	34°02'N, 109°27.5'W	2,495	0.161	5	5
Upper Nutrioso Creek	33°59.5'N, 109°29.9'W	2,389	0.220	5	13
Black River, West Fork, campground	33°47.4'N, 109°24.7'W	2,350	0.081	5	22
White River, North Fork	33°03.2'N, 109°4'W	2,334	0.050	15	23
Black River, Buffalo Crossing	33°45.8'N, 109°21.3'W	2,310	1.180	60	12
Little Colorado River, South Fork campground	34°04.8'N, 109°24.8'W	2,290	0.011	5	12
Little Colorado River, South Fork	34°05.3'N, 109°24.8'W	2,275	0.120	30	13
Upper Campbell Blue	33°44.8'N, 109°17.7'W	2,271	0.120	10	19
Black River, Wildcat Bridge	33°45.2'N, 109°27.2'W	2,125	0.190	35	9
Little Colorado River, Egar	34°06.7'N, 109°18.9'W	2,114	0.250	85	11
Linberg Springs, Flagstaff	35°08.8'N, 111°40.2'W	2,100	0.379	20	5
Double Springs Campground	34°56.5'N, 111°29.6'W	2,080	0.116	10	7
Little Colorado River, Springerville	34°08.8'N, 109°17.7'W	2,042	0.280	95	4
Blue River, Lazy Z Bridge	33°42.1'N, 109°04.2'W	1,894	0.330	50	7
East Turkey Creek, Chiricahua	31°54.8'N, 109°14.4'W	1,861	0.217	25	18
West Turkey Creek, Chiricahua	31°51.9'N, 109°21.5'W	1,748	0.140	5	12
Oak Creek, Sterling Springs	35°01.5'N, 111°44.4'W	1,828	0.150	5	23
Oak Creek, Pumphouse Wash	35°01.3'N, 111°44.2'W	1,755	0.150	5	30
Blue River, Lower Blue Crossing	33°38'N, 109°04'W	1,755	0.360	50	21
Oak Creek, Pine Flats	35°00'W 111°43.9'W	1,704	0.170	5	14
Cave Creek, North Fork	31°53.6'N, 109°12.9'W	1,705	0.014	25	8
Shake Springs, Chiricahua	32°01.3'N, 109°20.3'W	1,700	0.190	5	13
Oak Creek, West Fork confluence	34°59.3'N, 111°44.9'W	1,675	0.180	5	18
Oak Creek Spring	34°59.29'N, 111°44.9'W	1,674	0.180	5	16
Cave Creek, South Fork	31°52.5'N, 109°10.9'W	1,670	0.148	15	15
Cave Creek, Southwest Research Station	31°53.1'N, 109°12.4'W	1,646	0.310	20	12
Cave Creek, 1 km below Southwest Station	31°53'N, 109°12.3'W	1,642	0.310	20	8
Oak Creek, below West Fork	34°58.7'N, 111°44.8'W	1,639	0.180	5	18
Granite Creek, Prescott	34°32.9'N, 112°28.1'W	1,590	0.522	20	5
Upper Tonto Creek	34°20.4'N, 111°05.6'W	1,587	0.250	15	31
Willow Creek, Prescott	34°35.4'N, 112°29.9'W	1,569	0.363	25	2
Little Colorado River, Holbrook	34°52.4'N, 110°06.5'W	1,524	3.110	100	1
Kanab Creek, Kanab	37°06'N, 112°2'W	1,519	0.488	85	4
San Jan River	36°90'N, 109°95'W	1,440	0.750	85	7
Leslie Canyon Creek	31°36.8'N, 109°30.5'E	1,390	0.145	75	13

TABLE 1—Continued.

Stream site	Location	Elevation (m)	Specific conductance (mS/ cm)	Embedded- ness (%)	Species richness
Upper Fossil Creek Springs	34°25.5'N, 111°34.6'W	1,375	0.592	5	29
Oak Creek, Indian Gardens	34°54.6'N, 111°43.7'W	1,373	0.200	15	38
Fossil Creek, below Irving Plant	34°25'N, 111°34'W	1,350	0.600	15	11
Oak Creek, Sedona Bridge	34°51.8'N, 111°45.6'W	1,346	0.210	20	19
Oak Creek, Chavez Crossing	34°50'N, 111°46'W	1,283	0.263	20	13
Verde River, Paulden	39°51.9'N, 112°25.7'W	1,275	0.400	35	6
Willow Creek, near Kingman	34°10.9'N, 113°28.4'W	1,225	0.450	40	13
Wet Beaver Creek Campground	34°40.2'N, 111°48'W	1,220	0.530	15	6
San Pedro River	31°37'N, 110°10'W	1,205	0.500	70	3
Virgin River, North Fork, Springdale	37°10.6'N, 113°W	1,170	0.750	45	9
Gila River, Duncan	32°43.5'N, 109°06'W	1,126	1.032	85	10
Sycamore Creek, near Payson	34°18.5'N, 111°21.7'W	1,123	0.473	30	5
Red Tank Draw	33°40.7'N, 111°43.4'W	1,123	0.225	35	11
West Clear Creek, Bullpen campground	34°32.2'N, 111°41.4'W	1,115	0.260	70	6
Oak Creek, Page Springs	34°45.9'N, 111°53.5'W	1,092	0.402	40	25
Bubbling Springs Stream	37°46.4'N, 111°54.1'W	1,085	0.433	5	32
Fossil Creek near Verde River confluence	34°21'N, 111°43'W	1,075	0.592	45	10
Spring Creek, above Cottonwood	34°45.1'N, 111°54.9'W	1,060	0.521	35	13
Sheeps Head Irrigation Ditch	34°44.3'N, 111°55.6'W	1,050	0.860	45	5
Aravaipa Creek	32°54'N, 110°27'W	1,050	0.373	30	9
Verde River, Tuzigoot bridge	34°46.1'N, 112°02.4'W	1,038	0.470	80	10
Salt River, RK93, 3.5 km below US-60	33°47'N, 110°30'W	1,010	0.950	85	4
Salt River, RK79, 17.5 km below US-60	34°N, 110°W	990	0.950	85	9
Verde River, Camp Verde	34°33, 111°51.1'W	880	0.470	85	9
Colorado River, RK15	36°N, 112°W	930	0.850	90	3
Colorado River, RK50	36°N, 112°W	871	0.850	90	3
Colorado River, RK62	36°N, 112°W	850	0.850	90	4
Colorado River, Nankoweap confluence	36°17'N, 111°53'W	835	0.790	85	3
Little Colorado River, above confluence	36°14, W111°47'W	830	3.500	90	3
Salt River, RK51, 47 km below US-60	34°N, 110°W	820	0.950	85	5
Bright Angel Creek, above confluence	36°06.5'N, 112°06'W	736	0.330	20	9
Colorado River, Bright Angel Creek confluence	36°06.6'N, 112°06'W	734	0.850	95	3
Colorado River, RK290	36°11.5'N, 113°06'W	650	0.850	95	5
Colorado River, RK309	36°04'N, 113°20'W	640	0.850	95	2
Colorado River, RK346	35°53'N, 113°18'W	575	0.850	95	1
Hassayampa River Refuge	33°55.9'N, 112°41.6'W	597	0.640	25	6
Santa Maria River	34°22.2, 113°10.3'W	573	0.670	90	6
Colorado River, RK402	35°50.5, 113°40'W	400	0.853	95	1
Colorado River, Lake Havasu Marina	34°28.5'N, 114°21'W	250	0.980	95	2
Bill Williams River Refuge, Mineral Wash	34°15.3'N, 114°00'W	244	1.000	45	10
Achii Hanyo Fish Hatchery	34°02.2'N, 114°20.5'W	240	0.980	95	4
Colorado River, 3 km below Parker Dam	34°17'N, 114°10'W	240	0.980	90	6
Colorado River, 10 km below Parker Dam	34°15'N, 114°10.5'W	239	0.980	90	4
Gila River, near Yuma	32°45.6'N, 109°24.8'W	64	5.233	95	3

TABLE 2—Mean frequencies (% ± 1 SE) of selected caddisfly taxa, maximum percent at single site (maximum %), percent of streams where taxa occurred (% streams), range in elevation (m), and mean percent embeddedness for caddisfly species collected in the lower Colorado River basin, USA.

Species	Mean frequency (%)	Maximum (%)	% of streams	Range in elevation (m)	Mean embeddedness (%)
APATANIIDAE					
<i>Apatania arizona</i> (Apaari)	1.6 $\pm$ 1.2	43.0	4.3	1,587–2,080	13.3 $\pm$ 5.9
BRACHYCENTRIDAE					
<i>Brachycentrus americanus</i> (Braame)	0.2 $\pm$ 0.1	7.0	5.4	2,290–2,757	5.8 $\pm$ 0.8
<i>B. occidentalis</i> (Braocc)	0.2 $\pm$ 0.2	12.0	2.2	2,334–2,757	7.5 $\pm$ 2.5
<i>Micrasema bactro</i> (Micbac)	<0.01	<0.01	1.1	1,587	15
<i>M. onisca</i> (Miconi)	0.03 $\pm$ 0.02	1.0	2.2	1,587–2,490	10.0 $\pm$ 1.3
CALAMOCERATIDAE					
<i>Phylloicus aeneus</i> (Phyaen)	<0.01	0.01	3.3	1,275–1,748	20 $\pm$ 7.8
<i>P. mexicanus</i> (Phymex)	0.02 $\pm$ 0.02	1.0	3.2	1,275–1,748	20.0 $\pm$ 7.8
GLOSSOSOMATIDAE					
<i>Agapetus boulderensis</i> (Apabou)	0.04 $\pm$ 0.02	2.0	2.2	2,334–2,539	6.7 $\pm$ 1.7
<i>Culoptila cantha</i> (Culcan)	1.9 $\pm$ 1.1	5.0	10.8	240–1,977	45.0 $\pm$ 10.0
<i>C. kimminsi</i> (Culkim)	0.02 $\pm$ 0.02	2.0	2.2	1,546–1,618	12.5 $\pm$ 8.2
<i>C. moseleyi</i> (Culmos)	0.2 $\pm$ 0.01	9.0	8.6	1,587–2,600	6.9 $\pm$ 1.3
<i>C. thoracica</i> (Cultho)	0.003	<0.01	1.1	1,587	15
<i>Glossosoma ventrale</i> (Gloven)	0.5 $\pm$ 0.4	21.0	7.5	1,587–2,600	12.8 $\pm$ 6.5
<i>Protoptila erotica</i> (Proero)	0.4 $\pm$ 0.1	10.0	12.9	1,038–1,977	29.4 $\pm$ 8.4
HELICOPSYCHIDAE					
<i>Helicopsyche borealis</i> (Helbor)	2.3 $\pm$ 0.6	27.0	30.1	597–2,600	17.8 $\pm$ 2.7
<i>H. mexicana</i> (Helmex)	1.2 $\pm$ 0.4	24.0	18.3	1,030–2,490	26.0 $\pm$ 6.5
HYDROBIOSIDAE					
<i>Atopsyche sperryi</i> (Atospe)	1.9 $\pm$ 0.5	25.0	28.9	1,346–2,757	1.9 $\pm$ 3.2
<i>A. tripunctata</i> (Atotri)	0.05 $\pm$ 0.03	27.0	3.2	1,373–1,861	19.3 $\pm$ 5.7
HYDROPSYCHIDAE					
<i>Ceratopsyche cockerelli</i> (Cercoc)	0.02 $\pm$ 0.01	0.8	2.2	2,223–2,334	17.5 $\pm$ 7.6
<i>C. oslari</i> (Cerosl)	4.9 $\pm$ 0.1	48.0	32.3	640–2,757	10.3 $\pm$ 4.6
<i>C. venada</i> (Cerven)	1.1 $\pm$ 0.4	26.0	16.1	1,225–2,600	10.3 $\pm$ 2.6
<i>Cheumatopsyche arizonensis</i> (Cheari)	6.6 $\pm$ 1.5	64.0	43.0	1,000–2,525	21.2 $\pm$ 3.3
<i>C. enonis</i> (Cheeno)	2.0 $\pm$ 1.3	91.0	4.3	1,126–2,114	80.0 $\pm$ 5.4
<i>C. lasia</i> (Chelas)	<0.01	<0.01	2.2	2,125–2,490	40 $\pm$ 35.4
<i>C. pinula</i> (Chopin)	2.8 $\pm$ 1.1	74.0	15.1	1,030–2,525	32.4 $\pm$ 7.7
<i>Hydropsyche auricolor</i> (Hydaur)	2.5 $\pm$ 1.0	57.0	11.8	1,038–1,373	35.7 $\pm$ 3.3
<i>H. occidentalis</i> (Hydocc)	4.1 $\pm$ 1.2	81.0	30.1	975–2,600	33.9 $\pm$ 6.4
<i>Smicridea arizonensis</i> (Smiari)	<0.01	0.02	1.1	1,346	10
<i>S. dispar</i> (Smidis)	1.7 $\pm$ 0.8	60.0	12.9	1,030–1,755	39 $\pm$ 8.8
<i>S. fasciatella</i> (Smifas)	0.8 $\pm$ 0.4	32.0	6.5	64–1,373	52.2 $\pm$ 17.0
<i>S. signata</i> (Smisig)	0.8 $\pm$ 0.5	41.0	5.4	975–1,680	41.0 $\pm$ 15.5
HYDROPTILIDAE					
<i>Hydroptila ajax</i> (Hydaja)	5.3 $\pm$ 1.6	88.0	29.0	240–1,977	41.1 $\pm$ 5.7
<i>H. arctia</i> (Hydarc)	14.0 $\pm$ 2.0	100.0	59.1	400–2,600	31.1 $\pm$ 4.4
<i>H. hamata</i> (Hydham)	0.8 $\pm$ 0.3	14.0	18.3	1,046–2,757	21.4 $\pm$ 6.5
<i>H. icona</i> (Hydico)	2.8 $\pm$ 1.7	94.0	2.2	64–240	90.0 $\pm$ 0.0
<i>H. rono</i> (Hydron)	0.9 $\pm$ 0.4	29.0	15.1	1,275–2,596	14.1 $\pm$ 3.1
<i>Ithytrichia mexicana</i> (Ithmex)	1.5 $\pm$ 0.8	64.0	16.1	1,587–2,600	12.8 $\pm$ 5.0
<i>Leucotrichia pictipes</i> (Leupic)	0.01 $\pm$ 0.01	7.0	2.2	736–1,755	30.0 $\pm$ 11.8

TABLE 2—Continued.

Species	Mean frequency (%)	Maximum (%)	% of streams	Range in elevation (m)	Mean embeddedness (%)
<i>Mayatrichia acuna</i> (Mayacu)	0.04 ± 0.03	2.0	2.2	975–1,170	40.0 ± 0.0
<i>M. ayama</i> (Mayaya)	0.04 ± 0.03	2.0	2.2	975–1,030	60.0 ± 23.6
<i>Neotrichia halia</i> (Neohal)	0.1 ± 0.1	7.0	3.2	1,010–2,490	58.3 ± 27.2
<i>N. olorina</i> (Neolo)	<0.01	0.20	2.2	240	90.0 ± 0.0
<i>N. sonora</i> (Neoson)	2.6 ± 1.5	92.0	4.3	1,010–1,170	70.0 ± 15.3
<i>Ochrotrichia dactylophora</i> (Ochdac)	5.1 ± 1.5	74.0	36.6	650–2,596	23.7 ± 4.6
<i>O. ildria</i> (Ochild)	0.05 ± 0.02	1.0	7.5	1,373–2,596	8.3 ± 1.3
<i>O. logana</i> (Ochlog)	1.8 ± 1.1	74.0	3.2	640–835	91.7 ± 1.7
<i>O. lomenta</i> (Ochlom)	1.9 ± 1.3	96.0	6.5	736–2,600	33.8 ± 15.5
<i>O. (Metrichia) nigritta</i> (Ochnig)	0.4 ± 0.4	35.0	2.2	1,092–1,375	17.5 ± 14.7
<i>O. quadrispina</i> (Ochqua)	0.09 ± 0.04	3.0	7.5	1,346–2,080	12.6 ± 2.4
<i>O. spinulata</i> (Ochspi)	0.0072	<0.01	1.1	1,610	5.0 ± 5.0
<i>O. stylata</i> (Ochsty)	1.6 ± 0.6	2.0	19.4	573–2,290	38.5 ± 7.3
<i>O. tarsalis</i> Ochtar)	0.2 ± 0.1	5.0	8.6	975–1,755	36.3 ± 10.9
<i>Oxyethira aculea</i> (Oxyacu)	0.4 ± 0.3	21.0	3.2	573–2,757	38.3 ± 26.7
<i>O. arizona</i> (Oxyari)	1.4 ± 1.2	29.0	2.2	240	90.0 ± 0.0
<i>O. dualis</i> (Oxydua)	0.9 ± 0.4	25.0	11.8	573–1,977	38.2 ± 10.5
<i>O. pallida</i> (Oxypal)	0.3 ± 0.3	25.0	1.1	1,050	45.0 ± 45.0
<i>Zumatrichia notosa</i> (Zumnot)	0.7 ± 0.5	35.0	4.3	1,010–1,755	71.3 ± 10.7
LEPIDOSTOMATIDAE					
<i>Lepidostoma apache</i> (Lepapa)	<0.01	<0.01	1.1	1,861	25.0 ± 25.0
<i>L. bakeri</i> (Lepbak)	0.1 ± 0.1	6.0	3.2	1,546–1,865	18.3 ± 4.5
<i>L. knulli</i> (Lepknu)	0.2 ± 0.1	12.0	5.3	1,618–2,596	5.0 ± 1.0
<i>L. unicolor</i> (Lepuni)	0.8 ± 0.6	50.0	9.7	1,755–2,797	10.0 ± 3.8
LEPTOCERIDAE					
<i>Nectopsyche dorsalis</i> (Necdor)	0.3 ± 0.2	10.0	6.5	240–1,755	27.5 ± 13.2
<i>N. gracilis</i> (Necgra)	0.06 ± 0.06	5.0	1.1	1,275	30.0 ± 30.0
<i>N. stigmatica</i> (Necsti)	<0.01	<0.01	2.2	1,373–2,490	6.5 ± 1.8
<i>Oecetis arizonica</i> (Oecari)	0.3 ± 0.1	8.0	9.7	1,375–2,556	12.8 ± 3.5
<i>O. avara</i> (Oecava)	0.3 ± 0.2	11.0	11.8	1,085–2,490	30.0 ± 9.2
<i>O. disjuncta</i> (Oecdjs)	1.3 ± 0.5	23.0	19.4	1,618–2,797	10.3 ± 2.3
<i>O. inconspicua</i> (Oecinc)	0.07 ± 0.04	3.0	3.2	64–1,275	66.7 ± 31.0
<i>O. metlacensis</i> (Oecmet)	0.2 ± 0.2	16.0	1.1	2,334	10.0 ± 10.0
<i>Ylodes reuteri</i> (Yloreu)	0.05 ± 0.03	2.0	5.4	2,114–2,525	23.0 ± 13.8
LIMNephilidae					
<i>Amphicosmoecus canax</i> (Ampcan)	0.02 ± 0.02	2.0	2.2	2,539–2,757	5.0 ± 5.0
<i>Anabolia bimaculata</i> (Anabim)	0.9 ± 0.6	52.0	5.4	2,490–2,757	5.0 ± 5.0
<i>Clistronia maculata</i> (Climac)	0.8 ± 0.6	50.0	10.8	2,271–2,797	5.5 ± 0.5
<i>Hesperophylax magnus</i> (Hesmag)	0.4 ± 0.3	26.0	9.7	1,587–2,757	8.9 ± 2.3
<i>H. occidentalis</i> (Hesocc)	1.7 ± 1.0	89.0	16.1	1,894–2,757	11.0 ± 4.9
<i>Limnephilus apache</i> (Limapa)	0.3 ± 0.2	13.0	4.3	2,271–2,495	5.0 ± 5.0
<i>L. diversus</i> (Limdiv)	<0.01	0.13	2.2	2,495–2,539	5.0 ± 5.0
<i>L. frijole</i> (Limfri)	0.09 ± 0.08	7.0	2.2	1,587–2,271	10.0 ± 2.9
<i>L. granti</i> (Limgra)	0.3 ± 0.3	30.0	3.2	2,495–2,596	5.0 ± 5.0
<i>L. lithus</i> (Limlit)	0.5 ± 0.2	20.0	12.9	1,587–2,600	23.3 ± 9.1
<i>L. rothi</i> (Limrot)	0.03 ± 0.03	3.0	1.1	1,861	25.0 ± 25.0
<i>L. sperryi</i> (Limspe)	0.4 ± 0.4	33.0	3.2	2,500–2,757	5.0 ± 5.0
<i>L. tulatus</i> (Limtul)	<0.01	<0.01	1.1	1,587	15.0 ± 15.0
<i>Onocosmoecus unicolor</i> (Onouni)	0.1 ± 0.1	12.0	1.1	2,757	5.0 ± 5.0
<i>Psychoglypha schuhi</i> (Psych)	0.04 ± 0.02	2.0	4.3	2,495–2,757	5.0 ± 5.0
<i>P. subborealis</i> (Psub)	0.1 ± 0.1	10.0	4.3	2,125–2,757	22.5 ± 17.5

TABLE 2—Continued.

Species	Mean frequency (%)	Maximum (%)	% of streams	Range in elevation (m)	Mean embeddedness (%)
ODONTOCERIDAE					
<i>Marilia flexuosa</i> (Marfle)	0.4 ± 0.2	16.0	7.5	1,375–2,525	5.0 ± 1.0
<i>M. nobsca</i> (Marnob)	0.07 ± 0.06	6.0	2.2	1,085–1,755	5.0 ± 5.0
PHILOPOTAMIDAE					
<i>Chimarra angustipennis</i> (Chiang)	<0.01	<0.01	1.1	1,375	5.0 ± 5.0
<i>C. primula</i> (Chipri)	0.08 ± 0.05	3.0	4.3	1,373–1,861	17.0 ± 4.6
<i>C. ridleyi</i> (Chirid)	0.2 ± 0.2	29.0	3.2	1,346–1,680	9.3 ± 0.7
<i>C. utahensis</i> (Chiuta)	2.0 ± 0.5	29.0	24.7	640–2,490	30.1 ± 5.7
<i>Wormaldia arizonensis</i> (Worari)	0.9 ± 0.5	40.0	8.6	1,010–1,861	22.9 ± 9.5
POLYCENTROPODIDAE					
<i>Polycentropus arizonensis</i> (Polari)	1.5 ± 0.6	41.0	17.2	1,373–2,600	6.9 ± 1.3
<i>P. aztecus</i> (Polazt)	0.06 ± 0.04	4.0	3.3	2,490–2,600	5.0 ± 5.0
<i>P. gertschi</i> (Polger)	0.2 ± 0.1	8.0	7.8	1,123–1,700	21.4 ± 2.8
<i>P. halidus</i> (Polhal)	1.0 ± 0.3	20.0	24.4	575–2,556	22.5 ± 6.5
<i>P. variegatus</i> (Polvar)	<0.01	<0.01	1.1	1,587	15.0 ± 0.0
PSYCHOMYIIDAE					
<i>Psychomyia flavida</i> (Psyfla)	0.2 ± 0.1	10.0	4.4	1,283–2,525	18.3 ± 10.6
<i>Tinodes provo</i> (Tinpro)	0.09 ± 0.07	6.0	3.2	1,355–1,618	9.0 ± 3.9
RHYACOPHILIDAE					
<i>Rhyacophila angelita</i> (Rhyang)	0.01 ± 0.01	1.0	1.1	2,757	5.0 ± 0.0
<i>R. coloradensis</i> (Rhycol)	0.06 ± 0.04	3.0	4.3	1,755–2,541	11.3 ± 4.7
SERICOSTOMATIDAE					
<i>Gumaga griseola</i> (Gumgri)	1.7 ± 0.8	59.0	15.1	1,618–2,797	7.6 ± 2.6
UENOIDAE					
<i>Oligophlebodes minutus</i> (Olimin)	1.0 ± 0.7	54.0	6.5	2,495–2,797	5.8 ± 0.8

dedness (EMBI) indices showed a numerical relationship between caddisfly species and salinity and channel embeddedness, respectively. Caddisfly species with the lowest SCI values (i.e., *Brachycentrus americanus*, *Onocosmoecus unicolor*, *Psychoglypha subborealis*, and *Psychomyia flavida*) were most commonly associated with streams with low salinities (Table 3). Adjusted USEPA tolerance values for these species are 1 to 3 (Table 3). In contrast, species with the highest SCI values (i.e., *C. enonis*, *C. cantha*, *H. occidentalis*, *H. ajax*, *H. arctia*, and *Oecetis inconspicua*) reached highest numerical importance in streams with high salinities. Adjusted USEPA values for these species are 6 to 7 (Table 3).

In a similar manner, species with low EMBI values (i.e., *B. americanus*, *C. cantha*, *O. inconspicua*, and *P. subborealis*) were most commonly

associated with streams with low channel embeddedness, while species such as *C. enonis*, *H. occidentalis*, *H. arctia*, and *H. ajax* were most commonly associated with streams with high channel embeddedness. USEPA rankings for these species are provided in Table 3.

In general, SCI and EMBI values for genera were in less agreement with USEPA tolerance rankings than those for species. Only 7 (*Glossosoma*, *Limnephilus*, *Micrasema*, *Neotrichia*, *Oecetis*, and *Polycentropus*) out of the 42 genera collected in the lower basin were within 2 units of USEPA tolerance values cited for other regions.

Functional groups in the high-elevation sites (>2,000 m) of the lower basin included 47.9% collectors, 22.5% predators, 9% piercers, 6.6% shredders, and <2% grazers, whereas low-elevation and moderate-elevation groups were

TABLE 3—Specific conductance index (SCI) and channel embeddedness index (EMBI) values for selected caddisfly species in the lower Colorado River Basin, USA. Only species listed in Barbour et al. (1999) are used for comparison. Lower tolerance value indicates species is sensitive to specific conductance (salinity) and embeddedness. Values were adjusted to a scale of 0 to 10 for all caddisfly species collected according to Barbour et al. (1999) and compared with U. S. Environmental Protection Agency (USEPA) tolerance values for other regions. Species acronyms are provided in Table 2.

Species	SCI	Adjusted SCI tolerance value	USEPA tolerance	Species	EMBI	Adjusted EMBI tolerance value	USEPA tolerance value
Braame	1.89	1	1–2	Braame	0.93	1	1–2
Psysub	2.00	1	2	Culcan	1.24	2	0
Onouni	2.39	3	2	Psysub	1.26	2	2
Psyfla	2.49	3	1.9–3.3	Oecinc	1.41	3	8
Helbor	2.76	4	0–3.0	Helbor	1.61	4	0–3.0
Neohal	2.97	5	4	Psyfla	1.69	4	1.9–3.3
Cerosl	3.09	5	4	Onouni	1.77	4	2
Hydocc	3.18	6	4	Neohal	1.77	4	4
Cheeno	3.21	6	6	Cerosl	1.91	5	4
Oecinc	3.25	6	8	Hydocc	2.20	6	4
Hydarc	3.27	6	6	Hydarc	2.22	6	6
Culcan	3.41	7	0	Hydaja	2.35	7	6
Hydaja	3.51	7	6	Cheeno	2.71	8	6

dominated by piercers (73.3%, $SE \pm 33.3\%$) and collectors (13.1%, $SE \pm 6.7\%$).

DISCUSSION—Increased channel embeddedness and salinity, and reduced caddisfly species richness were closely related to land use in the lower basin. Channel embeddedness was the strongest determinant of caddisfly species richness, but embeddedness showed a significant interaction with specific conductance. These relationships are probably influenced by the heavy mining, grazing, and agriculture throughout many of the moderate-elevation to low-elevation streams, where catchment basins have been altered (Blinn and Poff, 2005). Channel siltation and secondary salinization are considered major stressors to stream biota in semiarid regions with high agricultural and municipal water demands (Williams, 1987; Laws, 2000).

Riparian vegetation was an important variable in caddisfly species richness. Fewer caddisfly species were collected at sites with the invasive salt-cedar than with native species of willow, alder, cottonwood, or sycamore. Salt-cedar might provide a lower quality habitat than native riparian species. Pomeroy et al. (2000) reported significantly slower breakdown of leaf

packs with salt-cedar compared to cottonwood leaf packs in the Colorado River.

Houghton (2004) recently reported 3 hydroptychids (*Cheumatopsyche campyla*, *Hydropsyche bidens*, and *Potamyia flava*) that were correlated with habitat disturbances in Minnesota. Caddisflies also have been used as indicators of water quality outside the USA. Berlin and Thiele (2002) classified watercourses in northeastern Germany on the basis of caddisfly distribution. Dohet (2002) also used caddisflies as bioindicators in streams in Luxembourg. Organic pollution, carbonate hardness, slope, and stream order best-explained differences in macroinvertebrate, especially caddisfly, composition among streams.

Barbour et al. (1999) provided tolerance ranges for 194 caddisfly species throughout the USA based on organic pollution, only 13 of which were collected in the lower basin. Nine of the 13 species were within one unit of the USEPA tolerance values. The close agreement with those species listed by Barbour et al. (1999) might be due to the fact that increases in channel embeddedness and salinity, especially the former, often complement organic enrichment in streams (Laws, 2000). The major discrepancies between rankings were with

C. cantha, where the adjusted SCI ranking in the lower basin was considerably higher than USEPA tolerances, and *O. inconspicua*, where the EMBI tolerances were considerably lower. Thus, the SCI and EMBI metrics estimated for species, but not genera, in the lower basin provide functional comparisons with published USEPA tolerance values.

We provide an updated list of caddisfly species in the lower Colorado River Basin that provides a baseline for future inventories on this group of aquatic insects in the region. We also provide information on the sensitivity of these species to channel embeddedness and specific conductance (salinity), 2 variables that frequently alter stream communities following disturbance in catchment basins (Laws, 2000).

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